

Micronutrient Whole Food Sources



Water-Soluble Vitamin	Whole-Food Sources
B₁ / Thiamin	Whole-grain cereal products, fortified breads, potatoes, legumes, nuts, pork, ham, liver
B₂ / Riboflavin	Dairy products (milk, yogurt, cheese), meat, liver, eggs, green leafy vegetables, beans
B₃ / Niacin	Meat, liver, poultry, fish, whole-grain cereal products, lentils, nuts; formed in the body from the essential amino acid tryptophan
B₅ / Pantothenic Acid	Liver, meat, dairy products (milk, yogurt, cheese), eggs, whole-grain cereal products, legumes, most vegetables
B₆ / Pyridoxine	Meat, liver, poultry, fish, whole-grain cereal products, potatoes, legumes, green leafy vegetables, dairy products (milk, yogurt, cheese), bananas, nuts
B₇ / Biotin	Meat, milk, egg yolk, whole-grain cereal products, legumes, most vegetables
B₉ / Folic Acid	Meat, liver, green leafy vegetables, whole-grain cereal products, potatoes, legumes, nuts, fruit
B₁₂ / Cobalamin	Meat, fish, shellfish, poultry, liver, eggs, dairy products (milk, yogurt, cheese), fortified breakfast cereals
Vitamin C / Ascorbic Acid	Citrus fruits, green leafy vegetables, broccoli, potatoes, peppers, strawberries
Choline*	Meat, poultry, fish, dairy products, and eggs, cruciferous vegetables and certain beans nuts, seeds, and whole grains

Adapted from Gropper & Smith (2022), Jeukendrup & Gleeson (2018, and Wallace et al. (2018) that may best reflect foods selected by physique athletes.

Fat-Soluble Vitamin and its Derivatives	Whole-Food Sources
Vitamin A (retinol, retinal, and retinoic acid) Provitamin A (Carotenoids)	Liver, fish, dairy products (milk, yogurt, cheese), eggs; are formed in the human body from Provitamin A (carotenoids) found in carrots, dark green leafy vegetables, tomatoes, oranges, sweet potato, carrots, spinach, butternut squash, greens, broccoli, and cantaloupe
Vitamin D (D2 – ergocalciferol and D3 – cholecalciferol) Provitamin (7-dehydrocholesterol)	Liver, fish, eggs, fortified dairy products, oils; formed by the action of sunlight on the skin
Vitamin E (Tocopherols and tocotrienols)	Liver, eggs, whole-grain cereal products, vegetable oils, seed oils
Vitamin K (Phylloquinones and menaquinones)	Liver, eggs, green leafy vegetables, cheese; formed in large intestine by bacteria

Adapted from Gropper & Smith (2022) and Jeukendrup & Gleeson (2018) that may best reflect foods selected by physique athletes.

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Macrominerals	Whole-Food Sources
Calcium / Ca^{2+}	Dairy products (milk, yogurt, cheese), egg yolk, beans, peas, dark green vegetables, canned sardines, clams, oysters, turnip and mustard greens, broccoli, legumes
Phosphorus / PO_4^{3-}	Meat, poultry, fish, eggs, dairy products (milk, yogurt, cheese), nuts, legumes, whole-grain products, soft drinks
Potassium / K^+	Fruits, vegetables, legumes, nuts, and dairy products (milk, yogurt, cheese), meat, fish, bread
Chloride / Cl^-	Table salt, seafood, meat, beans, canned foods, and eggs
Sodium / Na^+	Table salt, processed and snack foods, cured meats, seafood, condiments, milk, cheese, pickles, sauces, and bread
Magnesium / Mg^{2+}	Seafood, nuts, legumes, whole-grain products, leafy green vegetables, milk, yogurt

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Trace Minerals	Whole-Food Sources
Iron	Liver, red meat, eggs, clams, oysters, nuts, legumes, green leafy vegetables, dried fruits, broccoli, enriched/whole grains
Fluorine	Milk, egg yolk, seafood, drinking water
Zinc	Oysters, shellfish, beef, poultry, dairy products (milk, cheese, yogurt), liver, poultry, whole grains, legumes, vegetables, asparagus, spinach
Copper	Liver, shellfish, meat, fish, poultry, eggs, bran cereals, nuts, seeds, legumes, broccoli, bananas, avocados
Selenium	Liver, oysters, tuna, meat, poultry, fish, seafood, cereals, nuts from selenium-rich soil
Manganese	Wheat bran, legumes, hazelnuts, blueberries, pineapple, shellfish, poultry, meat, nuts, green leafy vegetables, bananas
Iodine	Iodized salt, saltwater seafood, seaweed, dairy products, liver, eggs, legumes, vegetables
Molybdenum	Liver, legumes, meat, poultry, fish, whole-grains, peas, beans
Chromium	Liver, meat, oysters, cheese, whole-grain products, asparagus, mushrooms, nuts, stainless steel cookware

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